

Heaven's Light is Our Guide

Rajshahi University of Engineering and Technology



Department of Mechatronics Engineering

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Name of Experiment : Study of Diode Clipper and Clamper Circuits
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Experiment No: 01

Name of Experiment: Study of Diode Clipper and Clamper Circuits

1.1 Objectives

1. To study the operation of positive and negative diode clipper circuits.
2. To study the operation of positive and negative diode clamper circuits.
3. To observe how DC bias and a coupling capacitor change the output waveform shape and level.
4. To compare experimental results with theory for waveform clipping and level shifting.

1.2 Theory

1.2.1 Clipper Circuits

Diode clipper circuits are used to limit the amplitude of an input waveform by removing one half of the signal above or below a reference level.[1] In a positive clipper, the diode conducts during the positive half-cycle and prevents the output voltage from rising above the diode threshold.[2] In a negative clipper, the diode conducts during the negative half-cycle and removes the negative-going portion of the waveform.[3]

These circuits are widely used in waveform shaping because they can protect sensitive electronic components from excessive voltage and can also help create desired pulse or square-like signals from sinusoidal inputs.[3] Their operation depends strongly on the diode polarity, the input signal polarity, and whether an external DC bias is present.[2]

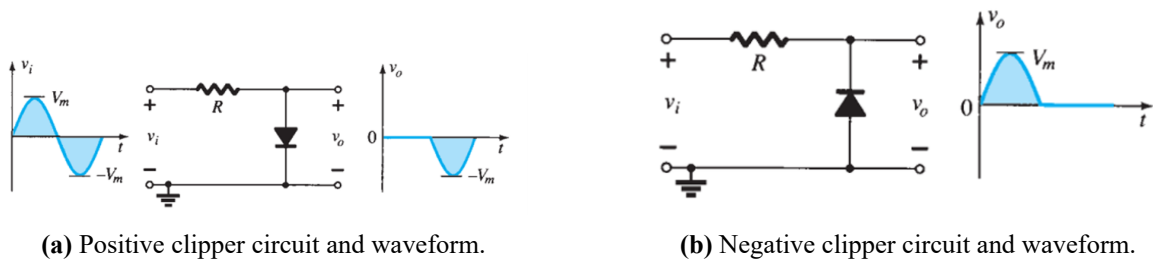


Figure 1.1: Unbiased diode clipper circuits and their output behavior.

When a DC bias source is connected in series with the diode, the clipping threshold shifts from zero to the bias voltage.[1] This allows the circuit to clip at a specified level rather than the nominal diode drop, as shown by the biased configurations below.[2]

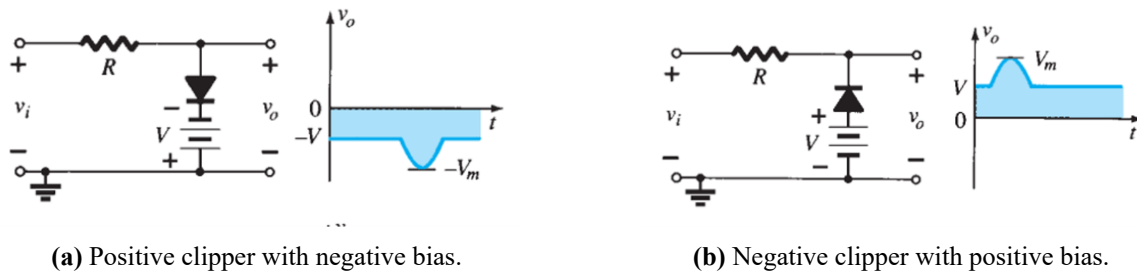


Figure 1.2: Biased diode clipper circuits with shifted clipping levels.

1.2.2 Clamper Circuits

Clamper circuits use a capacitor and diode to shift the DC level of an input waveform without removing one half of the signal.[1] A positive clamper raises the waveform baseline so the output stays above a reference level, while a negative clamper lowers the waveform baseline so the output stays below a reference level.[2]

In a typical clamper, the $10\ \mu\text{F}$ coupling capacitor charges during one half-cycle and then holds the charge, allowing the diode to clamp the waveform around the desired DC offset.[3] This makes clampers useful for level shifting and restoring signal reference levels in AC-coupled systems.[1]

The performance of a clamper depends on the capacitor charging time, the diode forward drop, and the source frequency, because these factors determine how accurately the output waveform is shifted to the intended level.[2] For this reason, clampers are commonly used in television receivers, communication circuits, and signal-conditioning applications where DC restoration is required.[3]

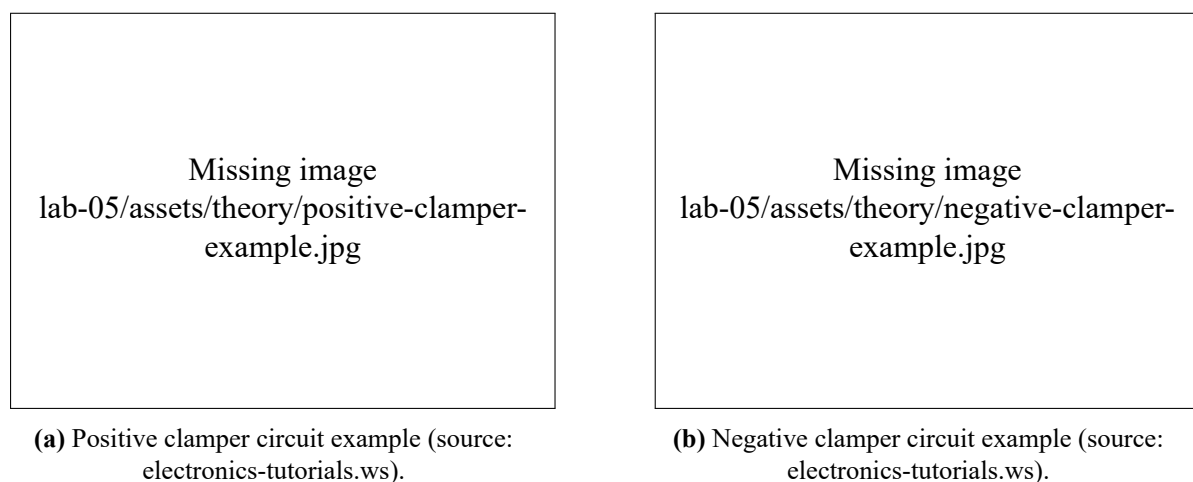


Figure 1.3: Unbiased clamper circuits that shift the waveform baseline.

Biasing a clamper with a DC source changes the clamped level by the bias amount, allowing the output waveform to be shifted upward or downward by a known offset.[2] A positive

clamper with positive bias clamps around a higher reference level, while a positive clamper with negative bias clamps around a lower one.[1] Similarly, a negative clamper with positive bias shifts the waveform downward, and a negative clamper with negative bias shifts it further downward.[3] In practical terms, a positive clamper biased positively raises the output waveform baseline, whereas a negative clamper biased negatively lowers it further, making the biased clamper useful for precise level shifting and waveform restoration.[2]

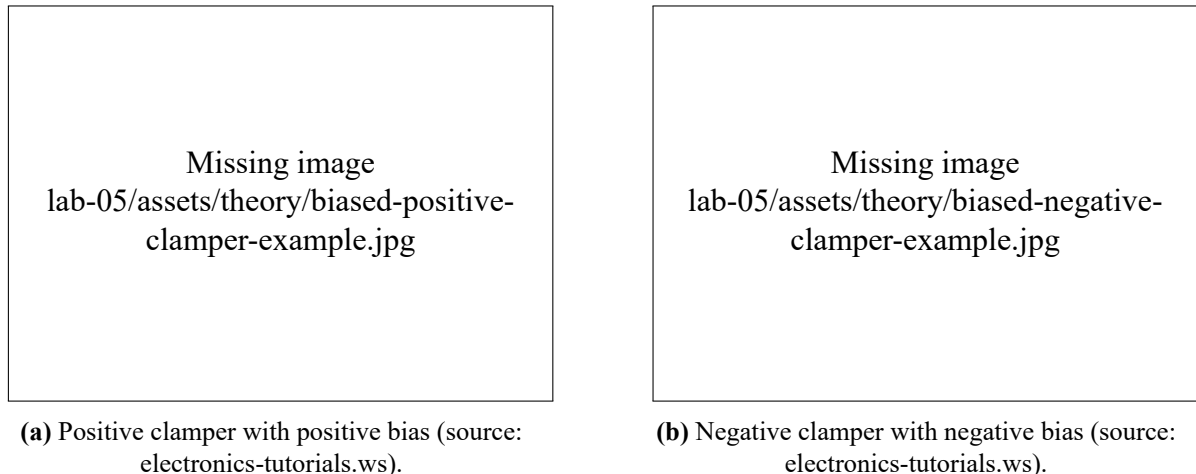


Figure 1.4: Biased clamper circuits showing shifted output reference levels.

These concepts are fundamental in electronic signal processing because they allow engineers to reshape signals without changing the original source waveform.[2] They are especially valuable in communication and instrumentation systems where precise level control is required.[1] By combining clipping and clamping principles, it is possible to design circuits that both limit amplitude and reposition the waveform baseline effectively.[3]

1.3 Apparatus Required

1. P-n junction diode: 1N4007 silicon diode.
2. Signal Generator
3. DC Regulated Power Supply
4. Resistor: 1 k Ω , 10 k Ω
5. Capacitor: 10 μ F
6. Oscilloscope
7. Oscilloscope chord (2 pieces)
8. Breadboard
9. Jumpers

1.4 Circuit Diagram

Clippers

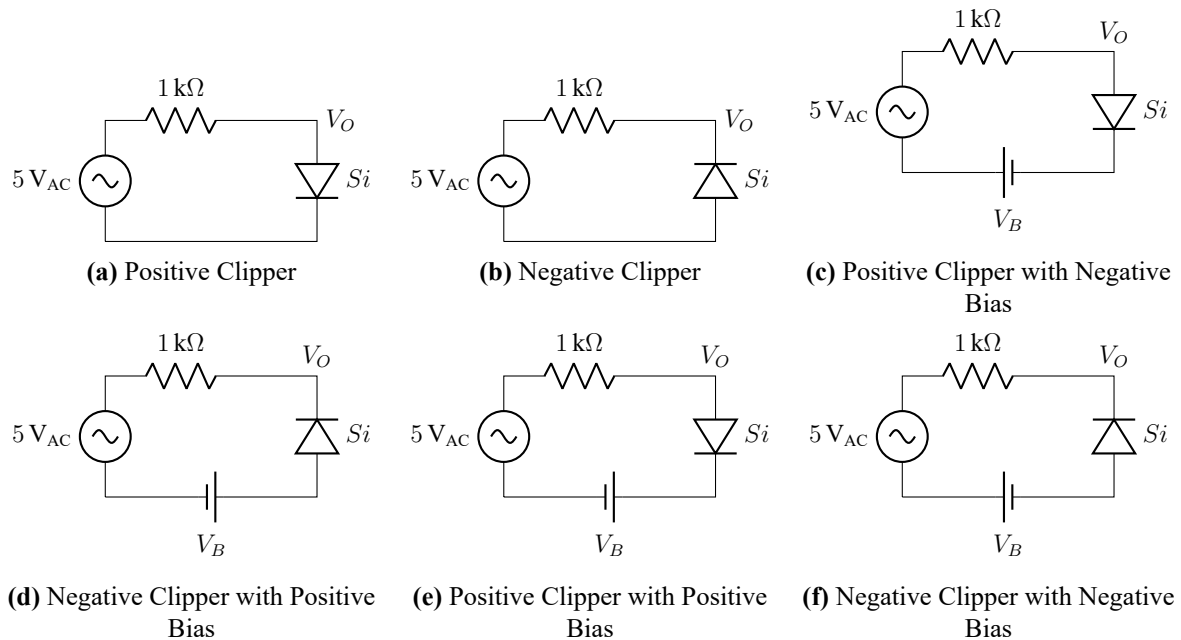


Figure 1.5: Six clipper circuit configurations studied in this experiment.

Clampers

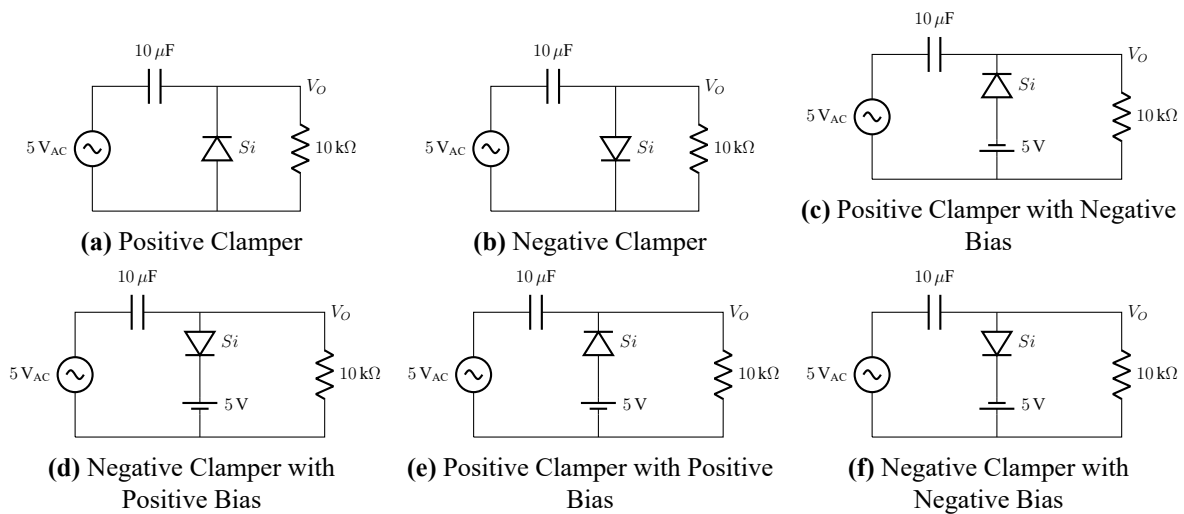


Figure 1.6: Six clamper circuit configurations studied in this experiment.

1.5 Result

Positive Clipper Circuit

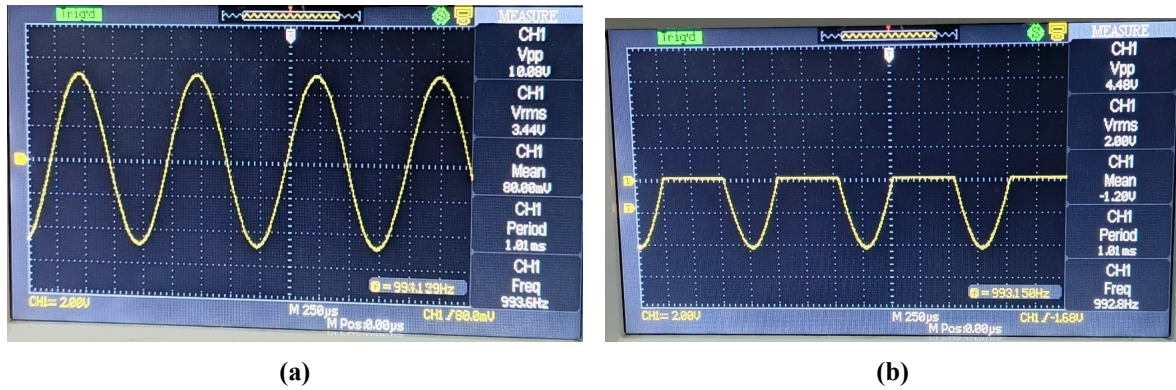


Figure 1.7: (a) Input and (b) Output Wave Shape for Positive Clipper (Sine Waves)

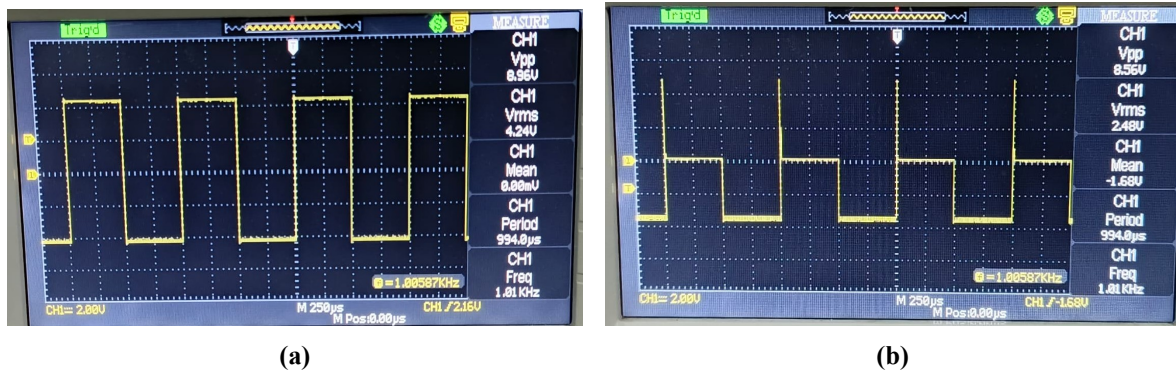


Figure 1.8: (a) Input and (b) Output Wave Shape for Positive Clipper (Square Waves)

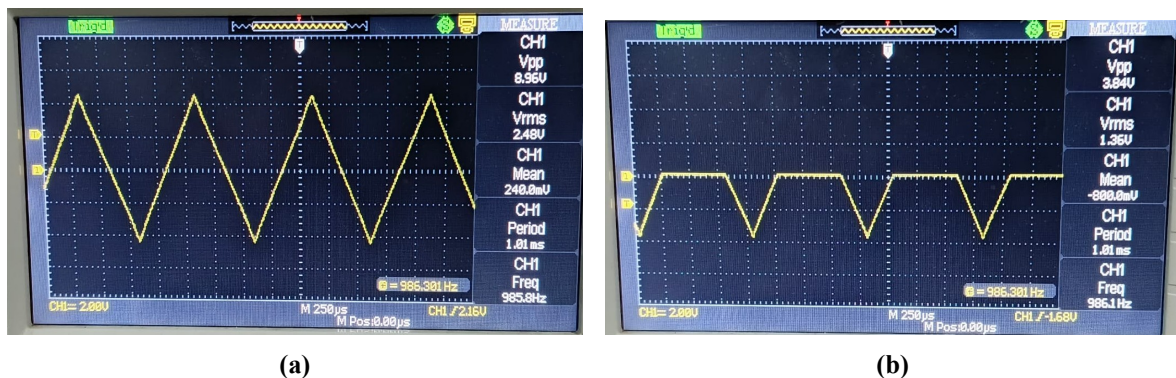


Figure 1.9: (a) Input and (b) Output Wave Shape for Positive Clipper (Triangular Waves)

Negative Clipper Circuit

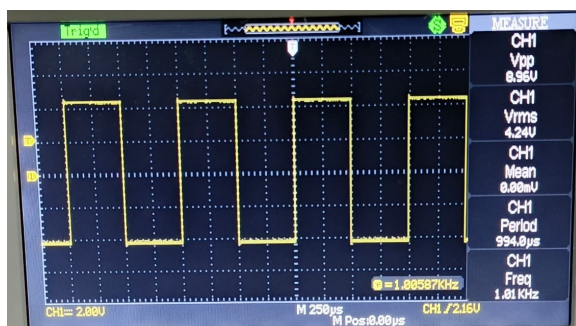


(a)



(b)

Figure 1.10: (a) Input and (b) Output Wave Shape for Negative Clipper (Sine Waves)

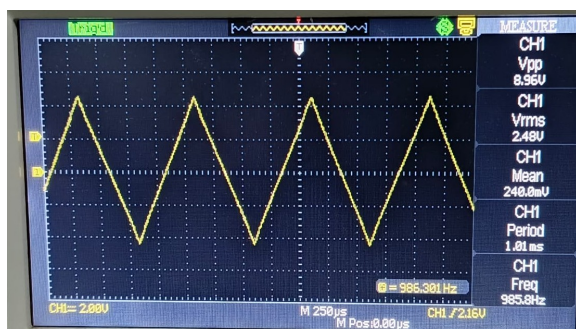


(a)



(b)

Figure 1.11: (a) Input and (b) Output Wave Shape for Negative Clipper (Square Waves)



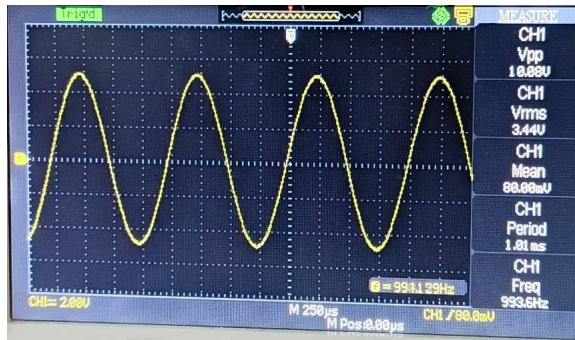
(a)



(b)

Figure 1.12: (a) Input and (b) Output Wave Shape for Negative Clipper (Triangular Waves)

Positive Clipper Positive Bias Circuit

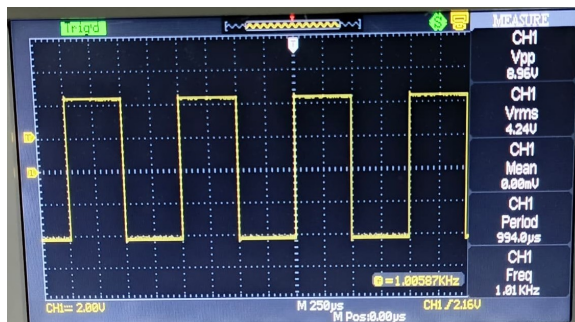


(a)

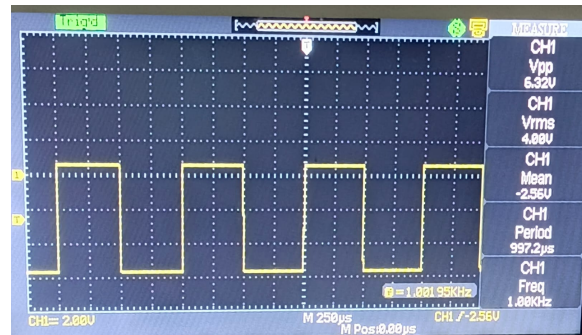


(b)

Figure 1.13: (a) Input and (b) Output Wave Shape for Positive Clipper Positive Bias (Sine Waves)

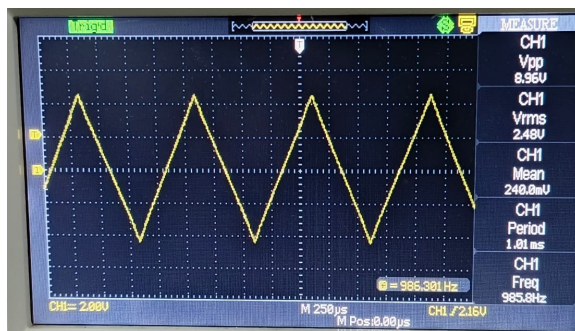


(a)

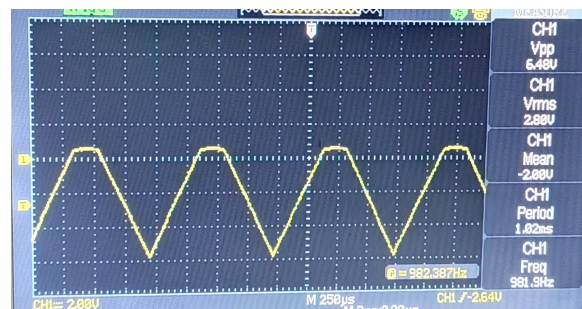


(b)

Figure 1.14: (a) Input and (b) Output Wave Shape for Positive Clipper Positive Bias (Square Waves)



(a)



(b)

Figure 1.15: (a) Input and (b) Output Wave Shape for Positive Clipper Positive Bias (Triangular Waves)

Negative Clipper Positive Bias Circuit

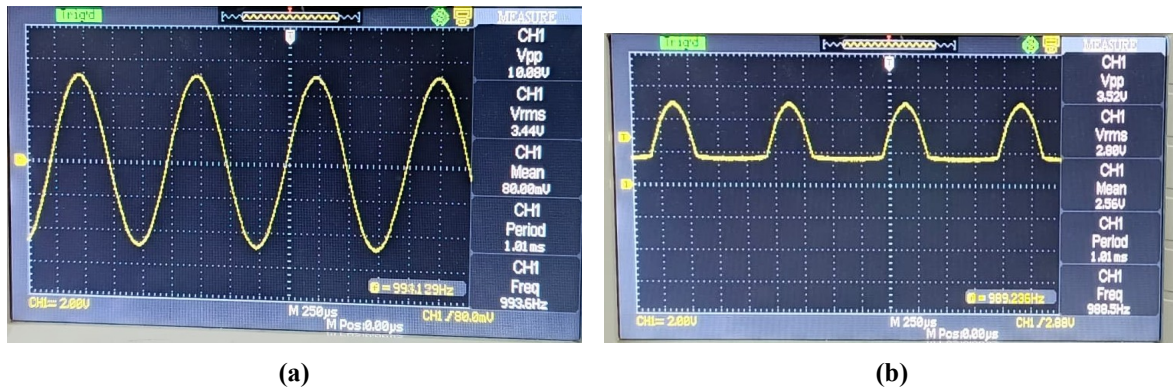


Figure 1.16: (a) Input and (b) Output Wave Shape for Negative Clipper Positive Bias (Sine Waves)

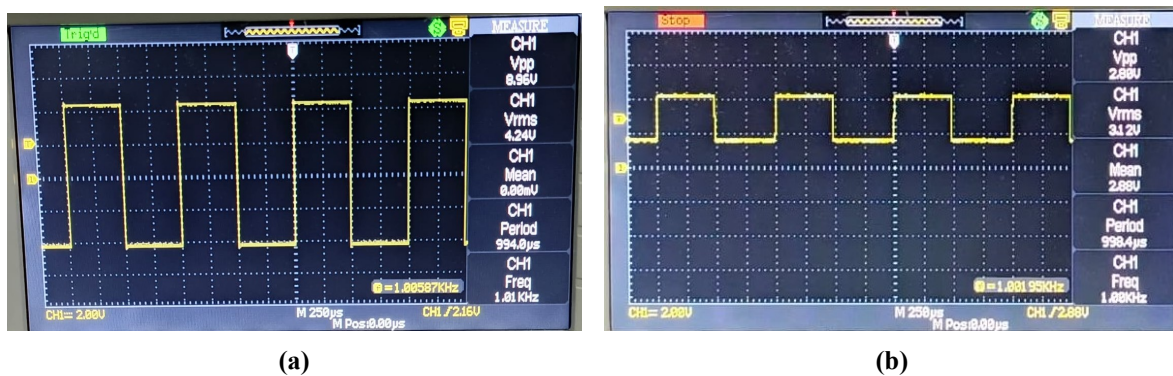


Figure 1.17: (a) Input and (b) Output Wave Shape for Negative Clipper Positive Bias (Square Waves)

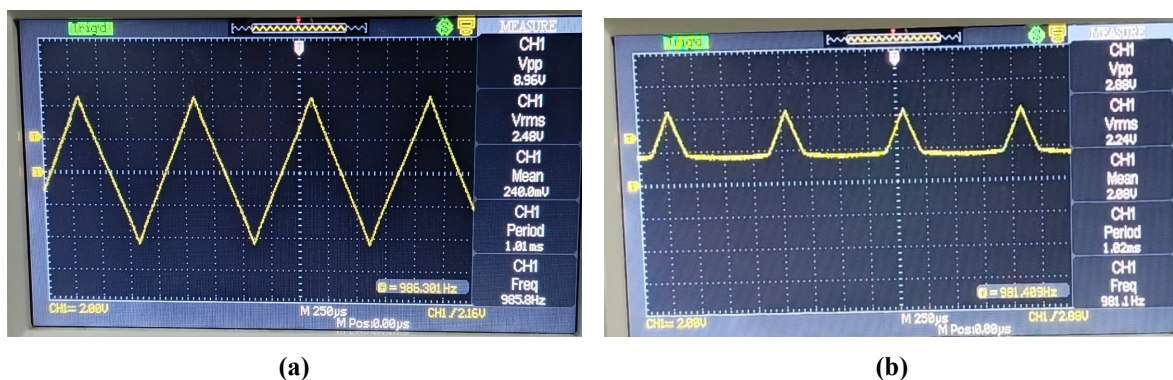


Figure 1.18: (a) Input and (b) Output Wave Shape for Negative Clipper Positive Bias (Triangular Waves)

Positive Clipper Negative Bias Circuit

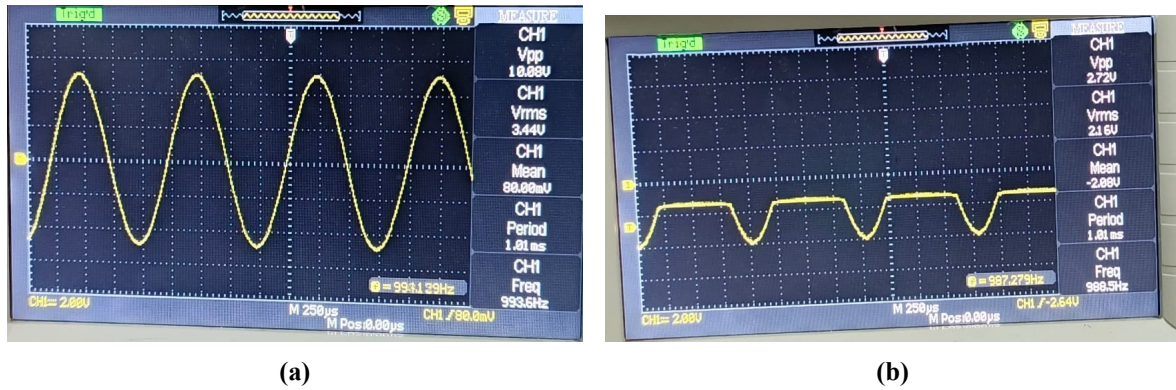


Figure 1.19: (a) Input and (b) Output Wave Shape for Positive Clipper Negative Bias (Sine Waves)

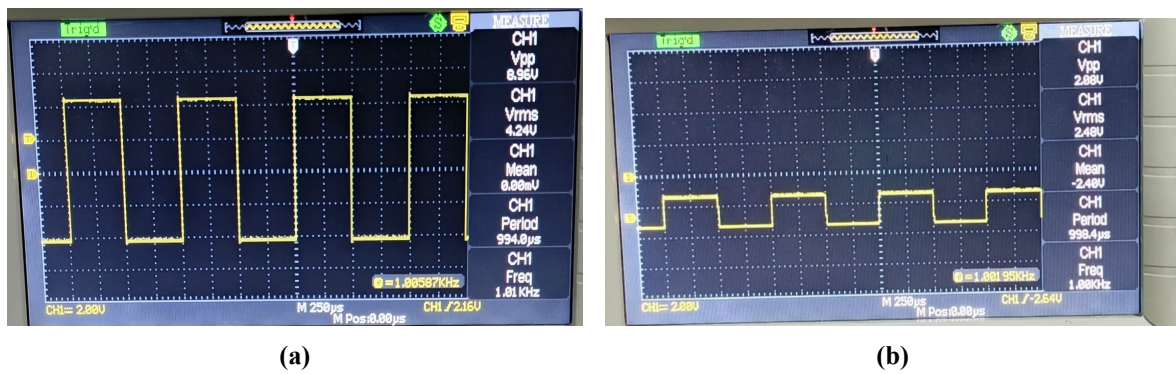


Figure 1.20: (a) Input and (b) Output Wave Shape for Positive Clipper Negative Bias (Square Waves)

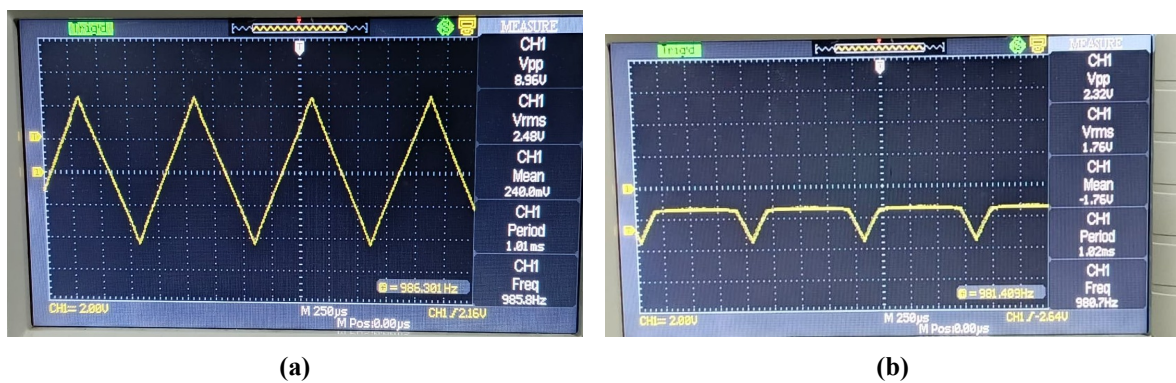
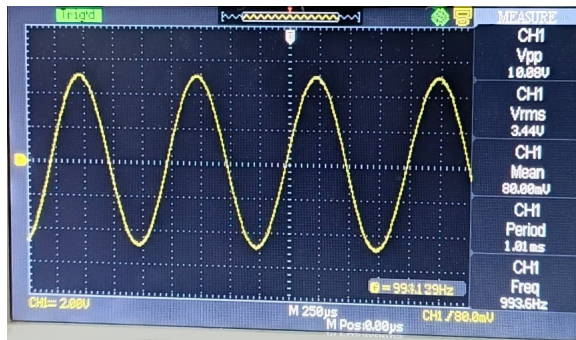


Figure 1.21: (a) Input and (b) Output Wave Shape for Positive Clipper Negative Bias (Triangular Waves)

Negative Clipper Negative Bias Circuit

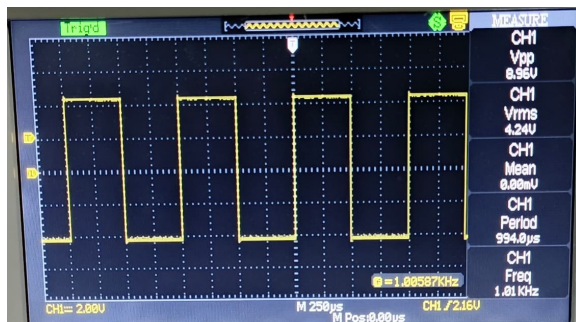


(a)

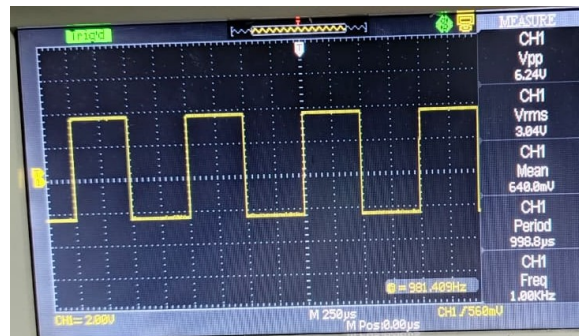


(b)

Figure 1.22: (a) Input and (b) Output Wave Shape for Negative Clipper Negative Bias (Sine Waves)

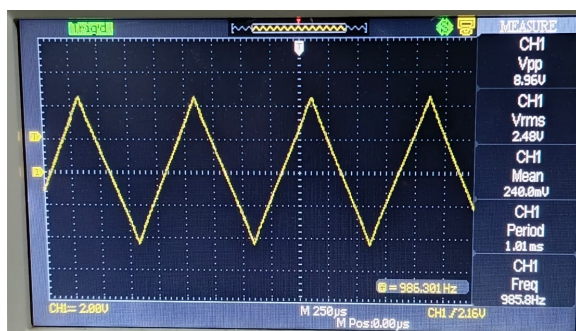


(a)



(b)

Figure 1.23: (a) Input and (b) Output Wave Shape for Negative Clipper Negative Bias (Square Waves)



(a)



(b)

Figure 1.24: (a) Input and (b) Output Wave Shape for Negative Clipper Negative Bias (Triangular Waves)

Positive Clamper Circuit

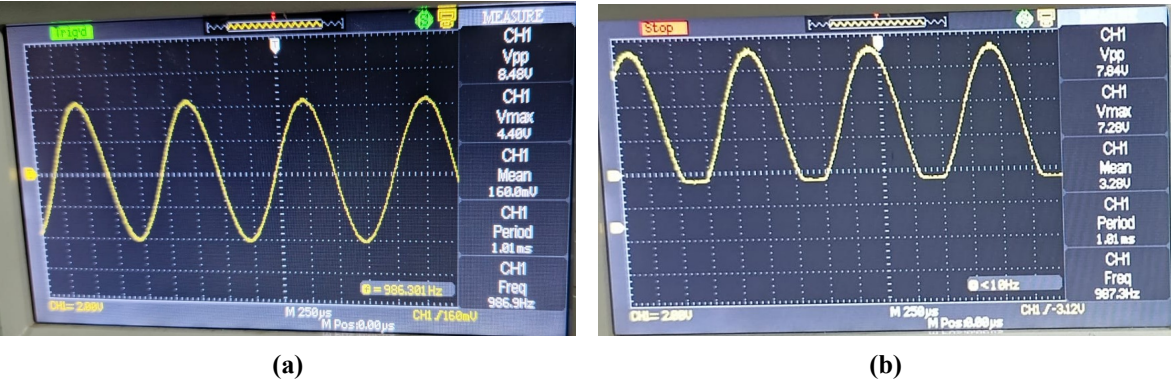


Figure 1.25: (a) Input and (b) Output Wave Shape for Positive Clamper (Sine Waves)

1.6 Simulaiton

Positive Clamper Circuit

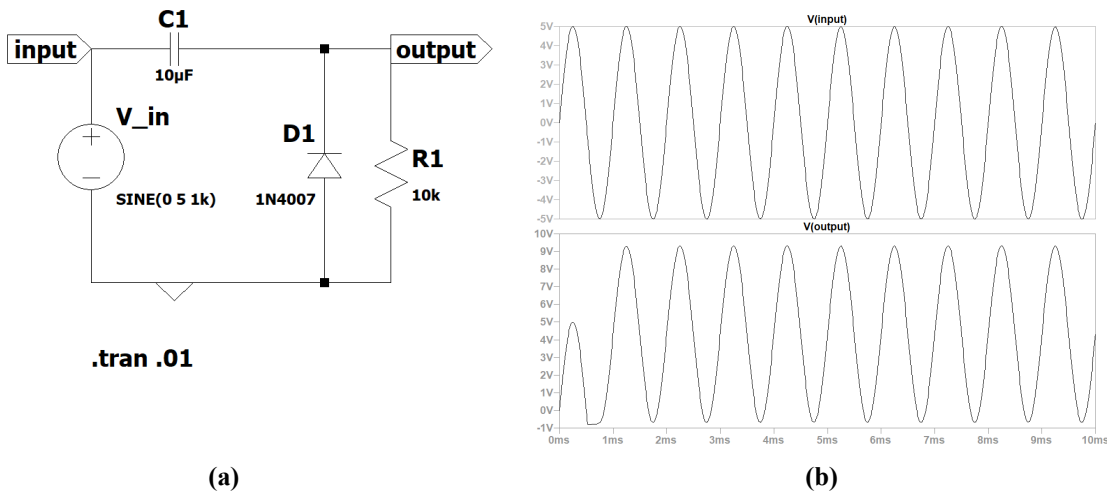
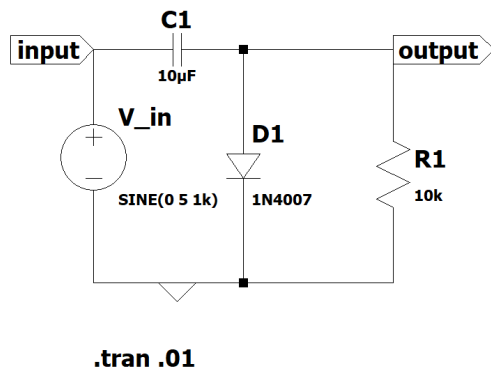
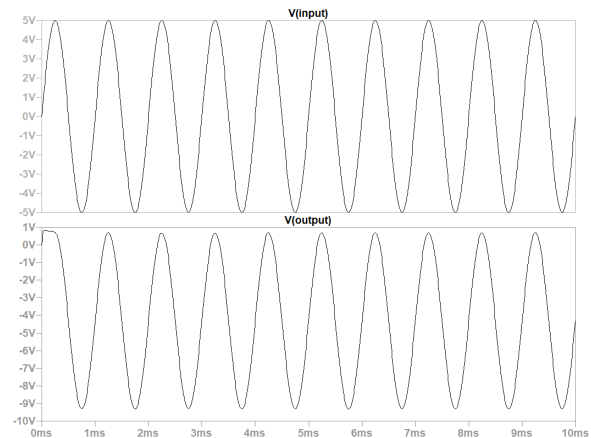


Figure 1.26: (a) Schematic and (b) Graph for Positive Clamper (Sine Waves)

Negative Clamper Circuit



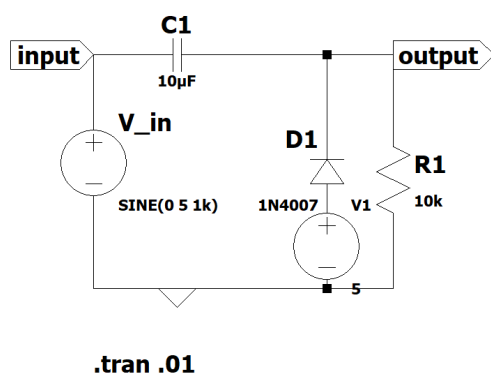
(a)



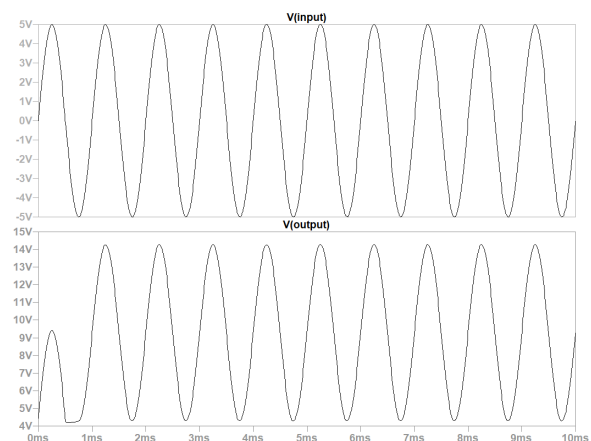
(b)

Figure 1.27: (a) Schematic and (b) Graph for Negative Clamper (Sine Waves)

Positive Clamper Positive Bias Circuit



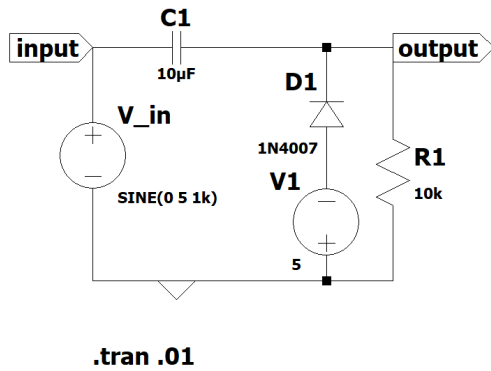
(a)



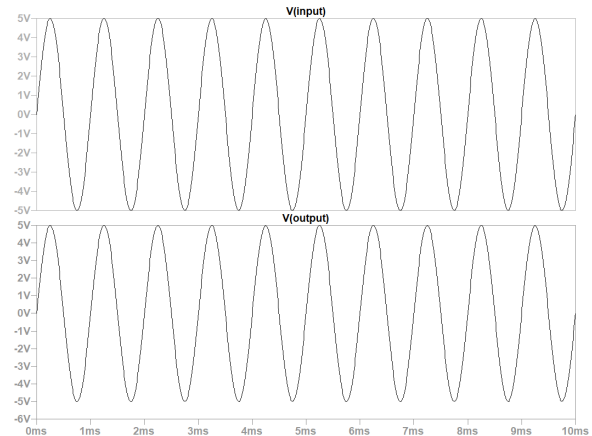
(b)

Figure 1.28: (a) Schematic and (b) Graph for Positive Clamper Positive Bias (Sine Waves)

Positive Clamper Negative Bias Circuit



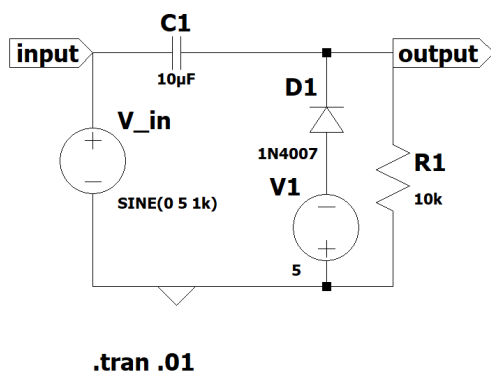
(a)



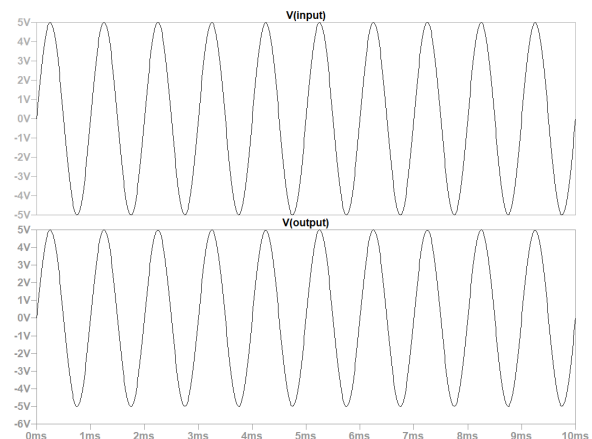
(b)

Figure 1.29: (a) Schematic and (b) Graph for Negative Clamper Postive Bias (Sine Waves)

Negative Clamper Positive Bias Circuit



(a)



(b)

Figure 1.30: (a) Schematic and (b) Graph for Negative Clamper Postive Bias (Sine Waves)

Negative Clamper Negative Bias Circuit

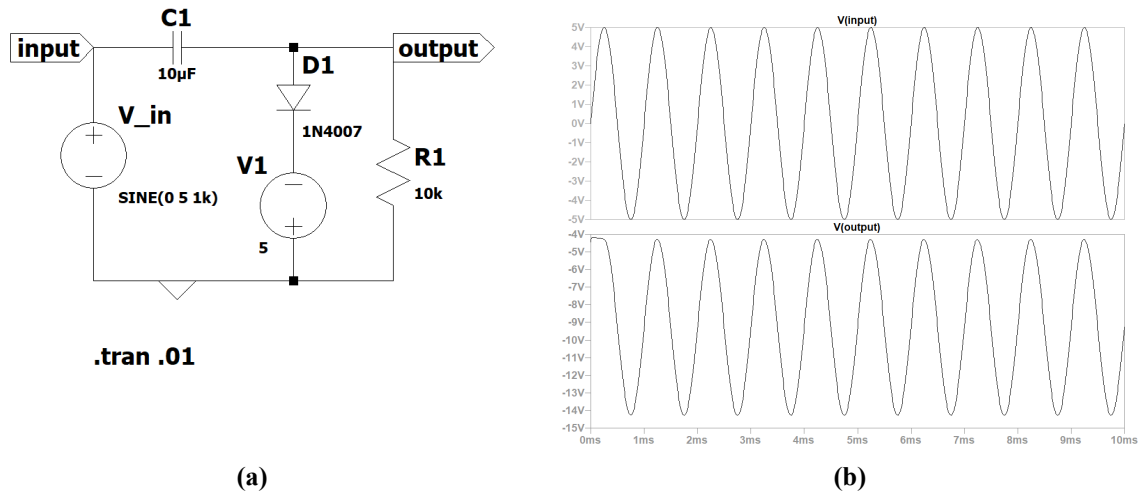


Figure 1.31: (a) Schematic and (b) Graph for Negative Clamper Negative Bias (Sine Waves)

1.7 Result and Discussion

The experimental result and discussion show that the diode clipper and clamper circuits behave consistently with theory: the clipper circuits remove the selected positive or negative half-cycle, while the clamper circuits shift the waveform baseline without eliminating an entire half-cycle. The observed output waveforms also reflect practical diode voltage drop and component tolerance, but the overall waveform shapes and bias-induced shifts matched the expected behavior for both biased and unbiased configurations.

1.8 Conclusion

The experiment confirmed that diode clippers remove the selected half-cycle of an input waveform while clampers shift the output level without removing the waveform shape, demonstrating the expected behavior of both signal-limiting and level-shifting circuits. Overall, the results provide a clear understanding of how diode-based waveform shaping circuits can be used in practical signal-conditioning and voltage-level adjustment applications.

References

- [1] R. L. Boylestad, *Electronic Devices and Circuit Theory*, 11th. Boston, MA: Pearson, 2012.

- [2] A. S. Sedra and K. C. Smith, *Microelectronic Circuits*, 7th. New York, NY: Oxford University Press, 2014.
- [3] P. Horowitz and W. Hill, *The Art of Electronics*, 3rd. Cambridge, UK: Cambridge University Press, 2015.